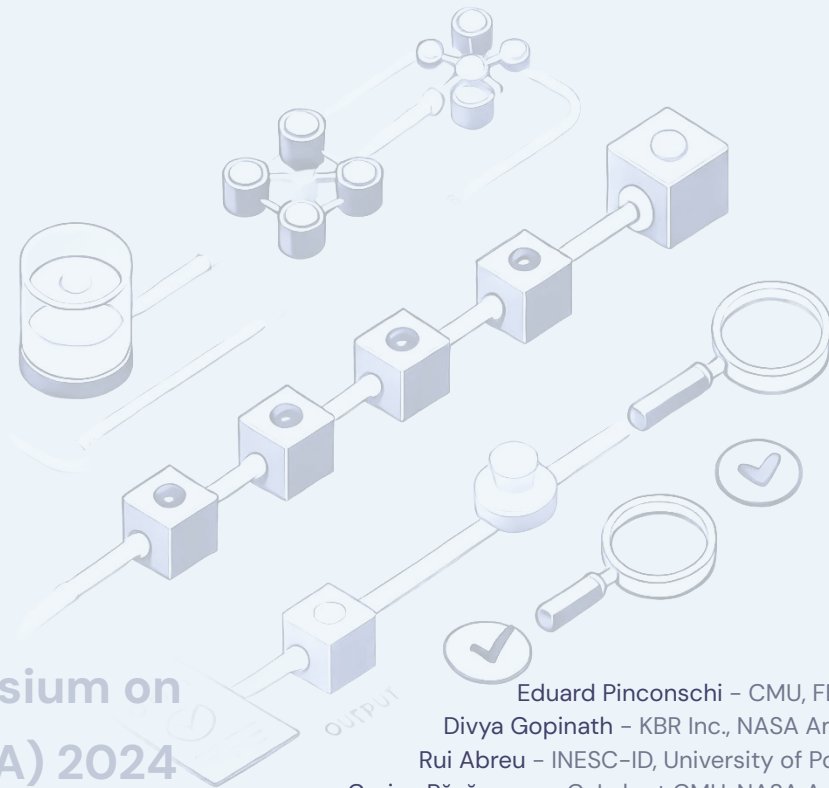


Evaluating Deep Neural Networks in Deployment

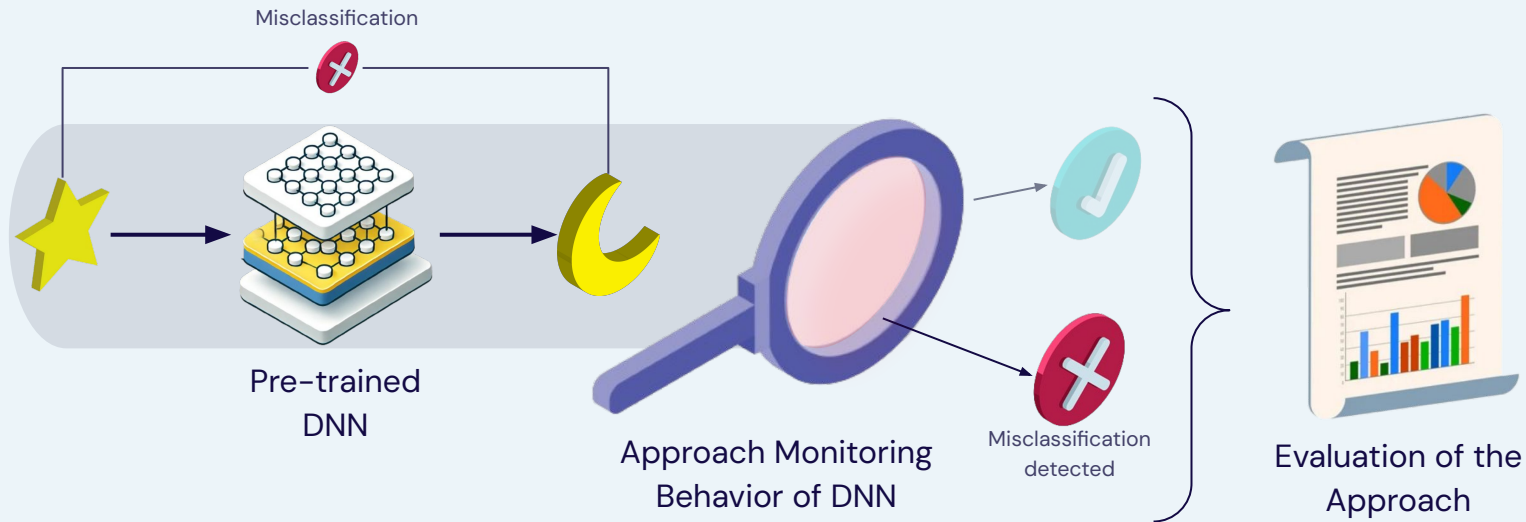
A Comparative Study
(Replicability Study)

33rd Edition of International Symposium on
Software Testing and Analysis (ISSTA) 2024



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Corina Păsăreanu – CyLab at CMU, NASA Ames

Evaluating Approaches for DNN Reliability

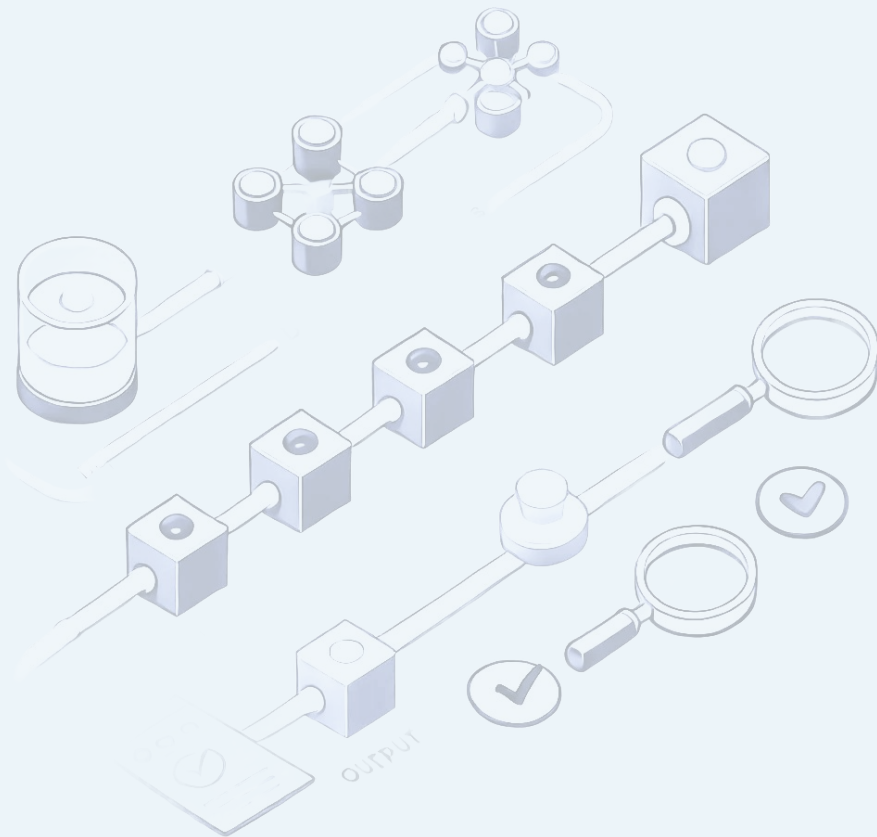


Outline

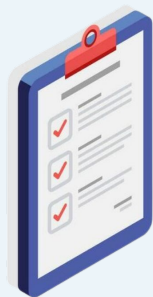
1. Introduction
 - a. Overview
 - b. Contributions
2. Approaches
 - a. Selection
 - b. Workflow
 - c. Algorithms
 - d. Challenges
3. Methods
 - a. TrustBench
 - b. Data preparation
 - c. Models
 - d. Tools
 - e. TrustDNN Framework
 - f. Metrics
4. Evaluation
 - a. Replication
 - b. Comparison
5. Conclusion
 - a. Lessons Learned
 - b. Final Remarks



Introduction



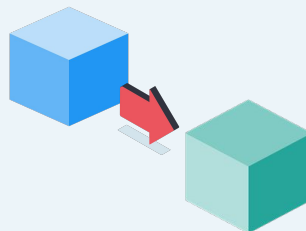
Overview



1. survey recent
approaches



2. reproduce on
own artifacts



3. reuse on
new domains



4. compare
performance

Contributions



identify SOTA
approaches



develop and
provide tools



replication study

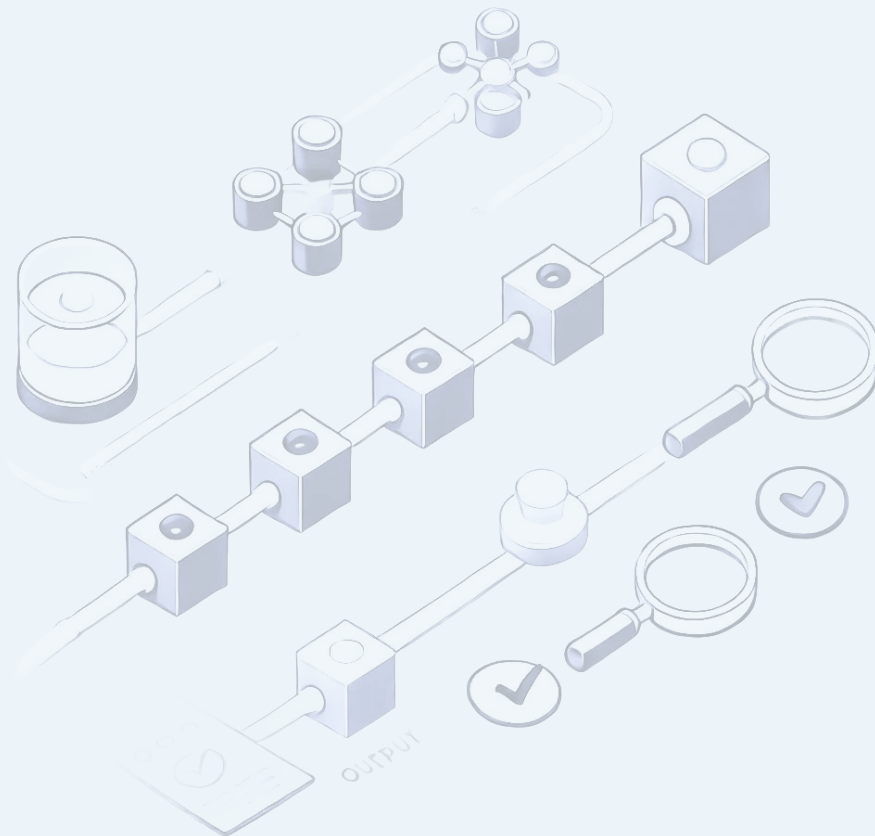


comparative study



replication package (artifacts and results)
<https://zenodo.org/doi/10.5281/zenodo.12803632>

Approaches



Selection

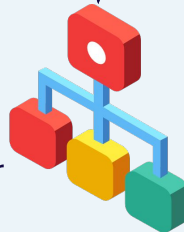


state-of-the-art;
within the last three years;
widely cited;



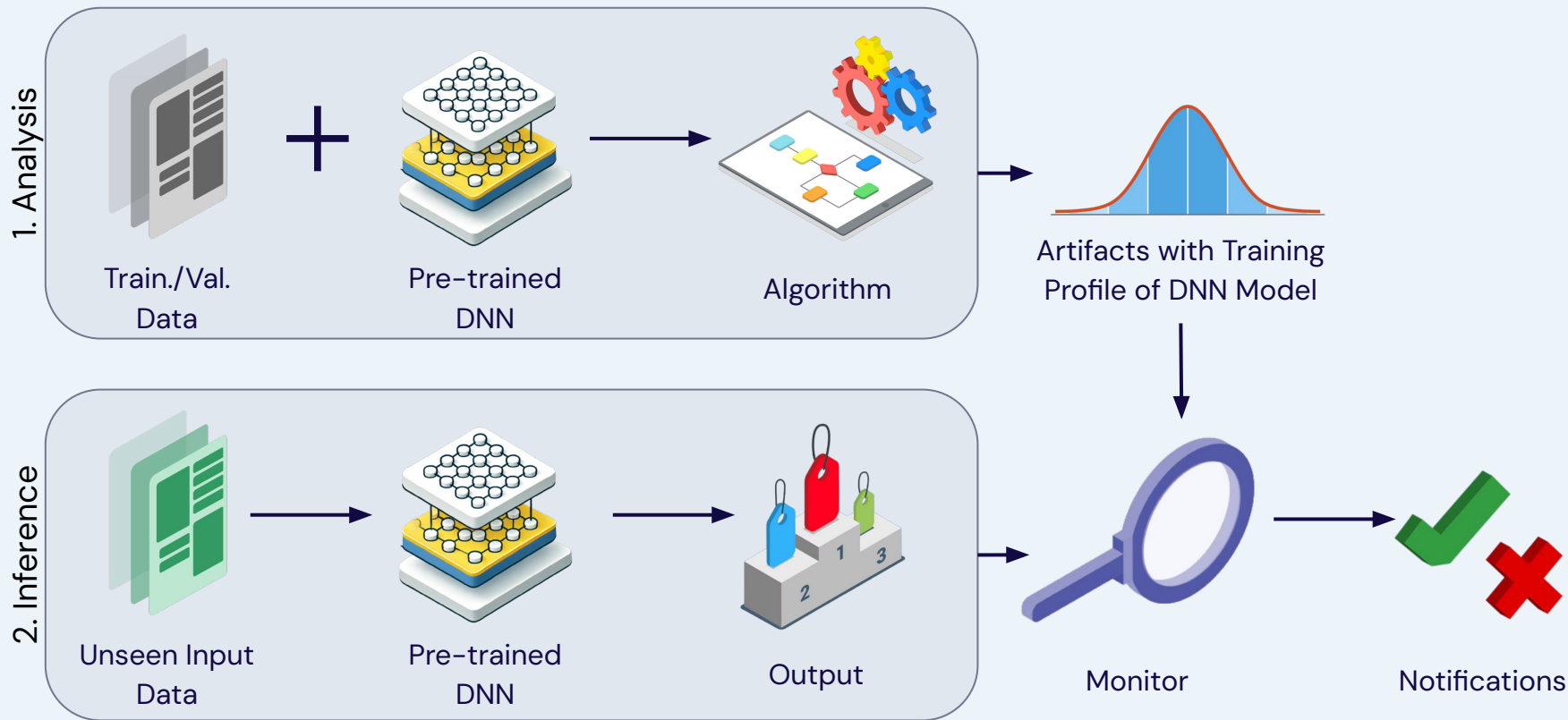
publicly available;
broad applicability;
under-evaluated;

- DeepInfer
- Prophecy
- SelfChecker

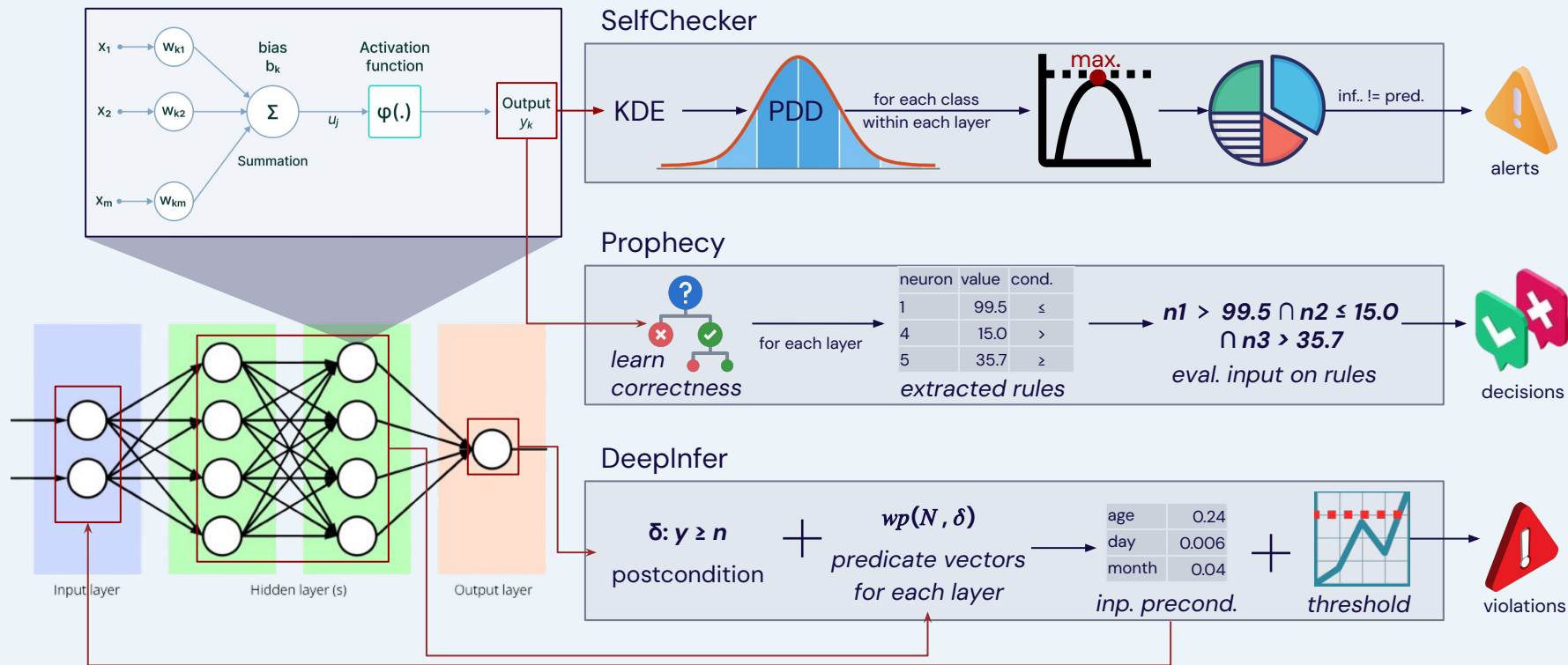


common workflow;
distinct algorithms;

Workflow



Algorithms



Challenges

Approaches



Implementation issues;



Lack of proper doc.;



Input domain mismatch;

Benchmark



Gap in input domain coverage;



Ad-hoc & non-uniform data preparation methods;

Evaluation



Lack of standard metrics;

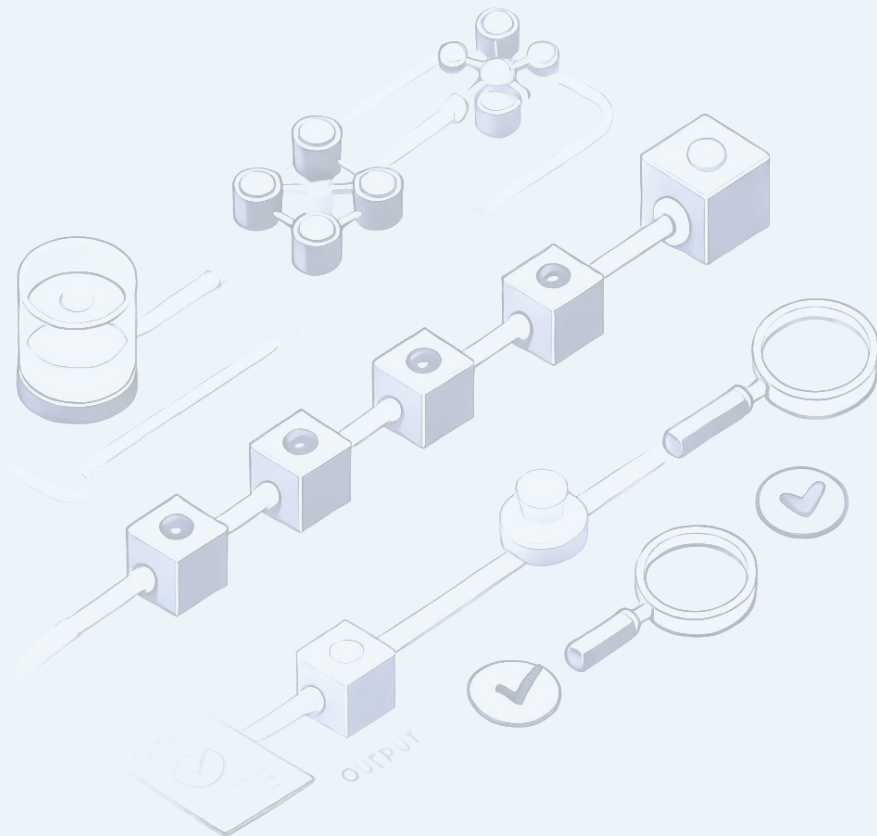


Closely coupled with code of algorithms;

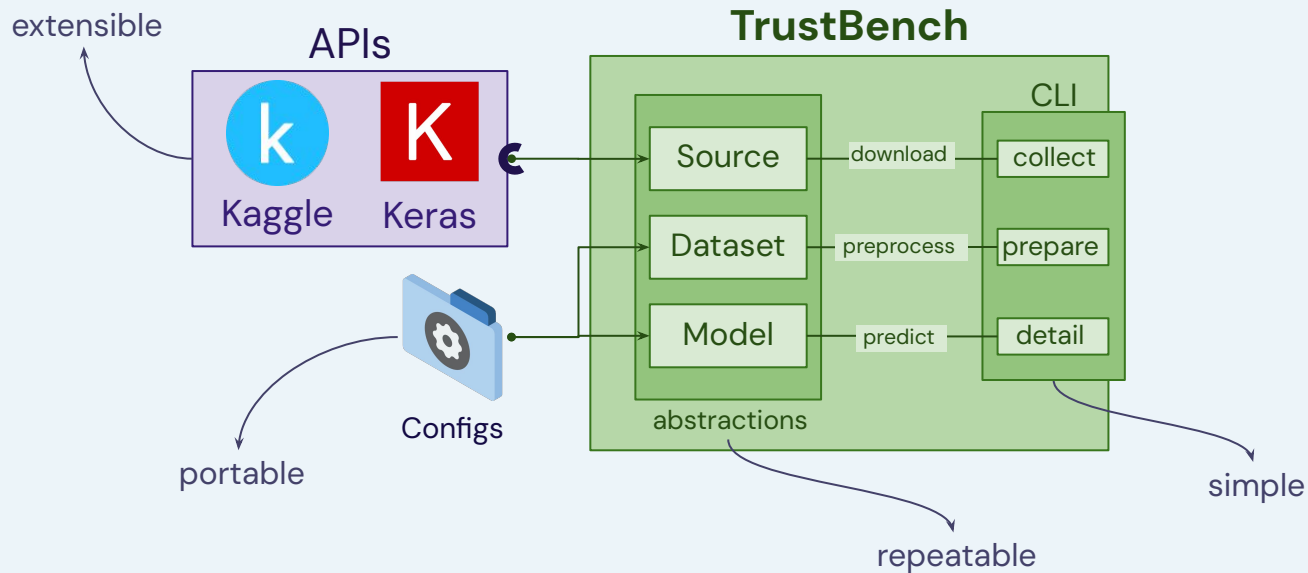


Inherent randomness in the training of models;

Methods



TrustBench – Tooling for Benchmark Creation



<https://github.com/epicosy/TrustBench>

Data preparation with TrustBench

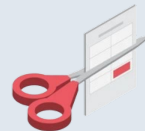
BM – Bank Marketing; GC – German Credit;
HP – House Price; PD – PIMA Diabetes; #L – Labels count;

	Dataset	#Samples				Input Size	#L
		Total	Train	Val.	Test		
DeepInfer replication datasets	BM	2116	NA	NA	2116	28	2
	GC	200	NA	NA	200	22	2
	HP	292	NA	NA	292	10	2
	PD	153	NA	NA	153	8	2
TrustBench-prepared datasets	BM	10578	8462	1058	1058	28	2
	GC	1000	800	100	100	22	2
	HP	1460	1168	146	146	10	2
	PD	768	614	77	77	8	2
Image dataset used by SelfChecker	CIFAR10	60000	40000	10000	10000	32*32*3	10



output class balancing;

one-hot encoded columns;



normalize numeric columns;

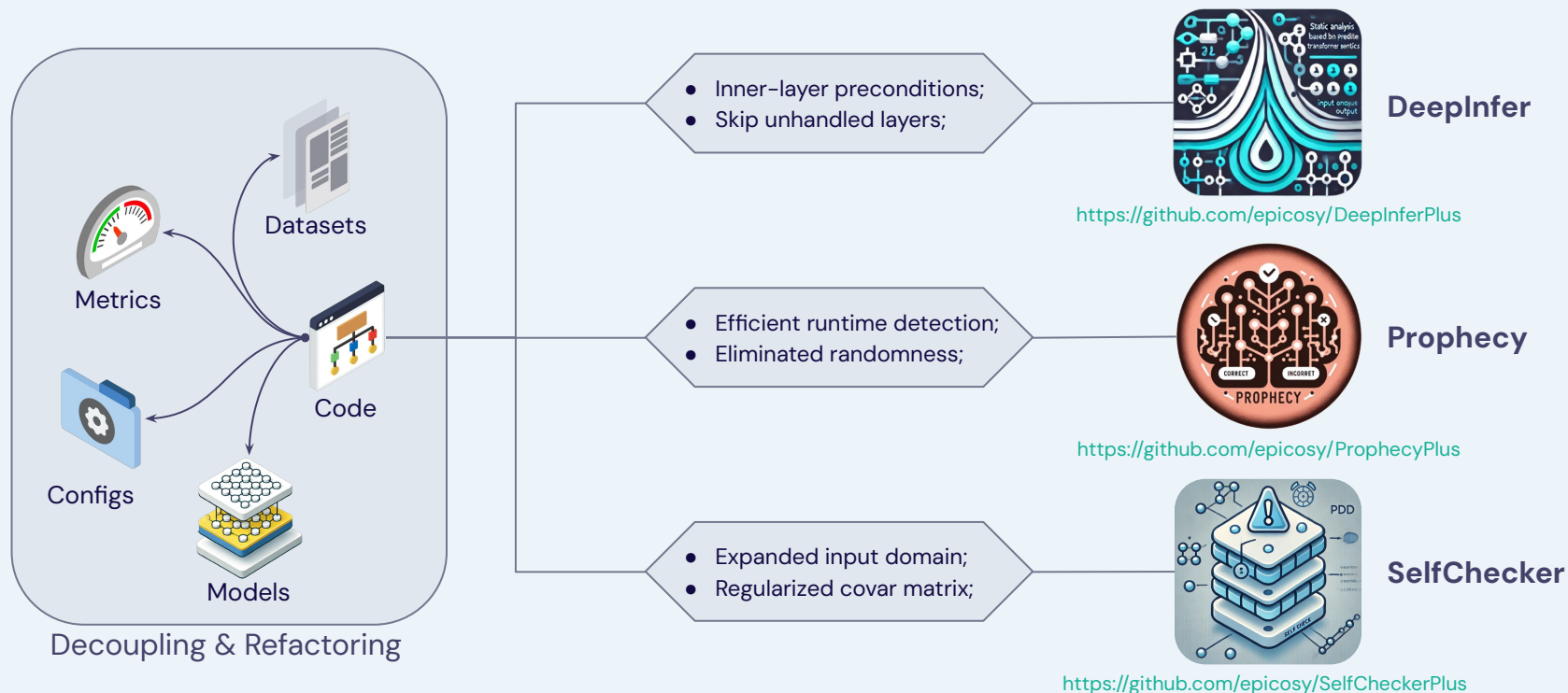
Description of the Models

Dataset	Model	Ground Truth		#L	#P	Acc.
		#C	#I			
BM	BM1	856	202	3	2913	80.91%
	BM2	863	195	3	1473	81.57%
	BM3	850	208	2	3001	80.34%
	BM4	846	212	4	24551	79.96%
	BM5	856	202	3	879	80.91%
	BM6	854	204	3	361	80.72%
	BM7	861	197	3	6081	81.38%
	BM8	868	190	6	4641	82.04%
	BM9	857	201	3	627	81.0%
	BM10	520	538	3	627	49.15%
	BM11	520	538	3	627	49.15%
	BM12	869	189	4	1439	82.14%
HP	HP1	124	22	3	273	84.93%
	HP2	122	24	3	273	83.56%
	HP3	74	72	3	273	50.68%
	HP4	128	18	4	383	87.67%

Dataset	Model	Ground Truth		#L	#P	Acc.
		#C	#I			
GC	GC1	67	33	2	1201	67.0%
	GC2	67	33	2	2401	67.0%
	GC3	67	33	2	217	67.0%
	GC4	67	33	3	171	67.0%
	GC5	67	33	6	4257	67.0%
	GC6	67	33	3	2295	67.0%
	GC7	67	33	3	2295	67.0%
	GC8	67	33	3	2295	67.0%
	GC9	67	33	4	2801	67.0%
PD	PD1	61	16	3	221	79.22%
	PD2	45	32	3	221	58.44%
	PD3	46	31	3	221	59.74%
	PD4	62	15	4	293	80.52%
CIFAR10	ConvNet	8045	1955	25	2.9M	80.45%

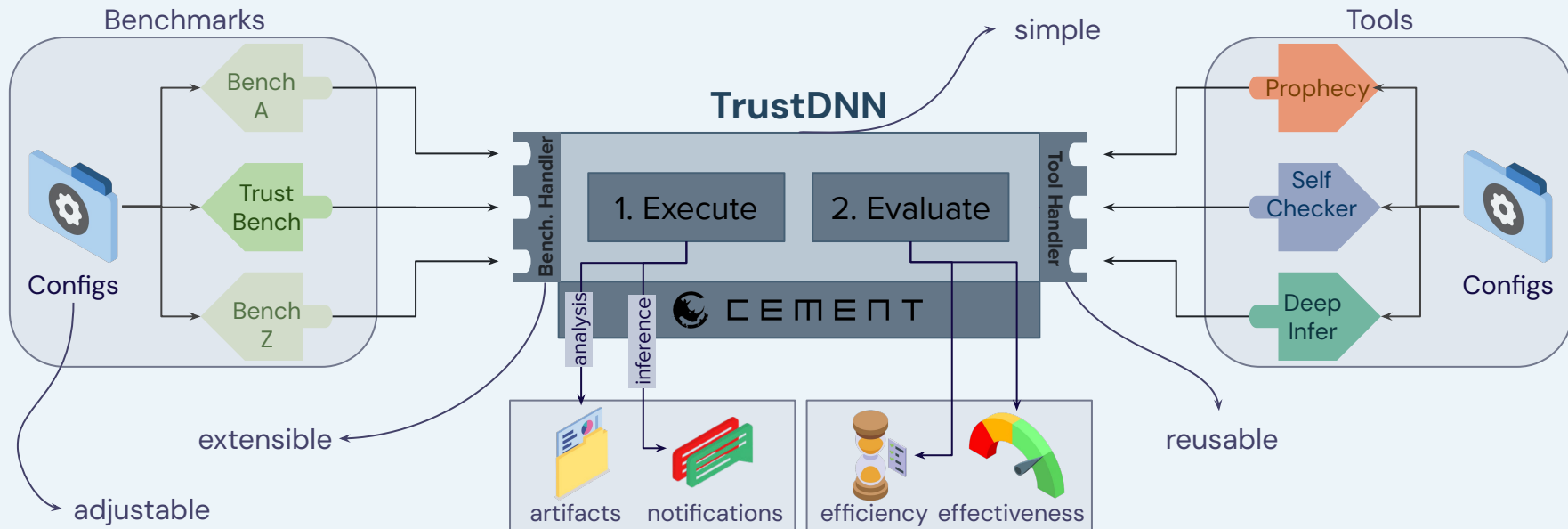
#C – Num. of Correct; #I – Num. of Incorrect; #L – Num. of Layers;
 #P– Num. of Trainable Parameters;

From Approach-Specific Code to Practical Tools



*The images of the tools are custom representations created to distinguish between them and not their official logos.

TrustDNN – Evaluation Framework



<https://github.com/epicosy/TrustDNN>

Metrics – Evaluating Misclassification with MCC

Conventional Metrics

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

$$\text{TPR (Recall)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

used in previous studies

offer valuable insights

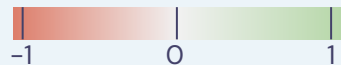
sensitive to class swapping

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}$$

considers all categories

considers dist. of elements

correlation between predictions/outcomes



misaligned

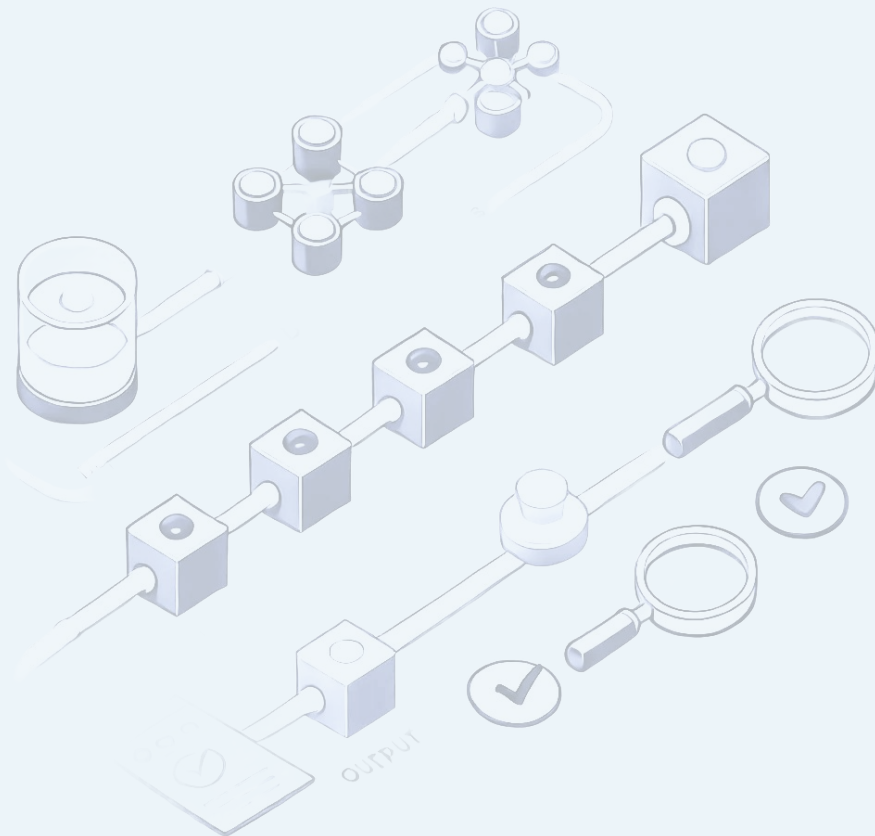


random



aligned

Evaluation



Research Questions

Replicability Analysis

- RQ1: For DeepInfer and SelfChecker, to what extent can the results be reproduced on their own artifacts (datasets and models)?
- RQ2: For the considered approaches, can the results be obtained on artifacts other than their own?

Comparative Analysis

- RQ3: How do the considered approaches compare in terms of effectiveness?
- RQ4: How do the considered approaches compare in terms of time and memory consumption?

Replication Results

RQ.1 – Can be reproduced on own artifacts?

	<i>Are we able to run?</i>	<i>Did we get the same results?</i>
DeepInfer	Yes	Partially (variations of setup)
SelfChecker	Yes	Yes (slight differences)

*Prophecy – was not considered before for the problem at hand

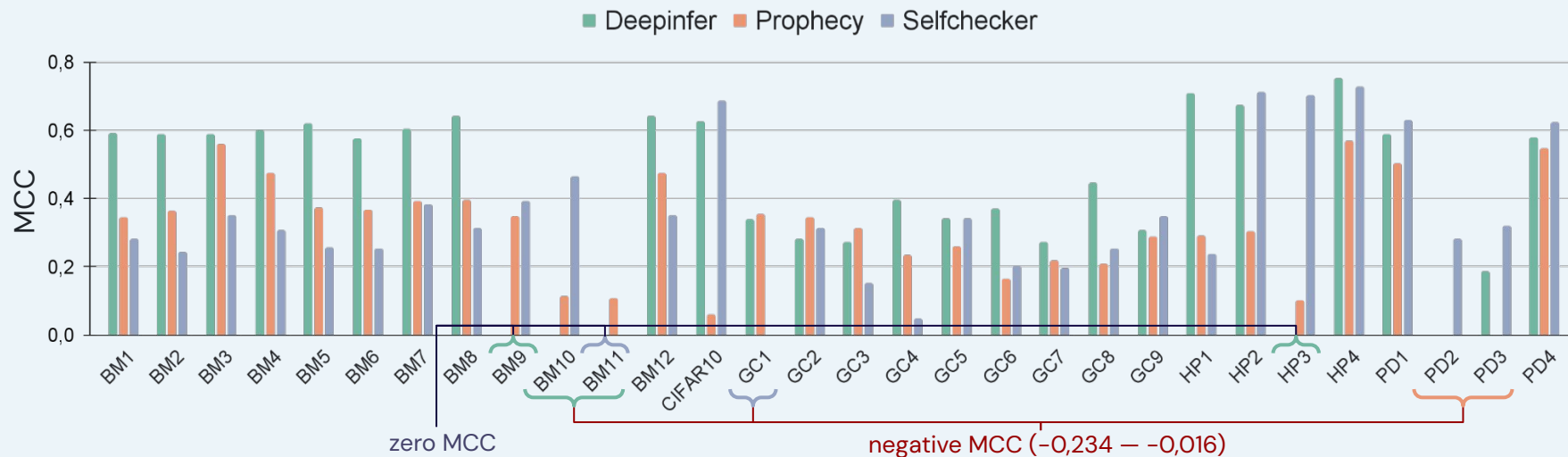
Observation 1: Successful but difficult to reproduce.

RQ.2 – Can run on artifacts other than their own?

	<i>Tabular Data</i>	<i>Image Data</i>
DeepInfer	—	No (cannot handle format)
SelfChecker	No (linear algebra error)	—
Prophecy	Yes	Yes

Observation 2: Unsuccessful; difficult replication on different artifacts; **Successful** with extended versions

Comparison Results – Effectiveness



Tabular data:

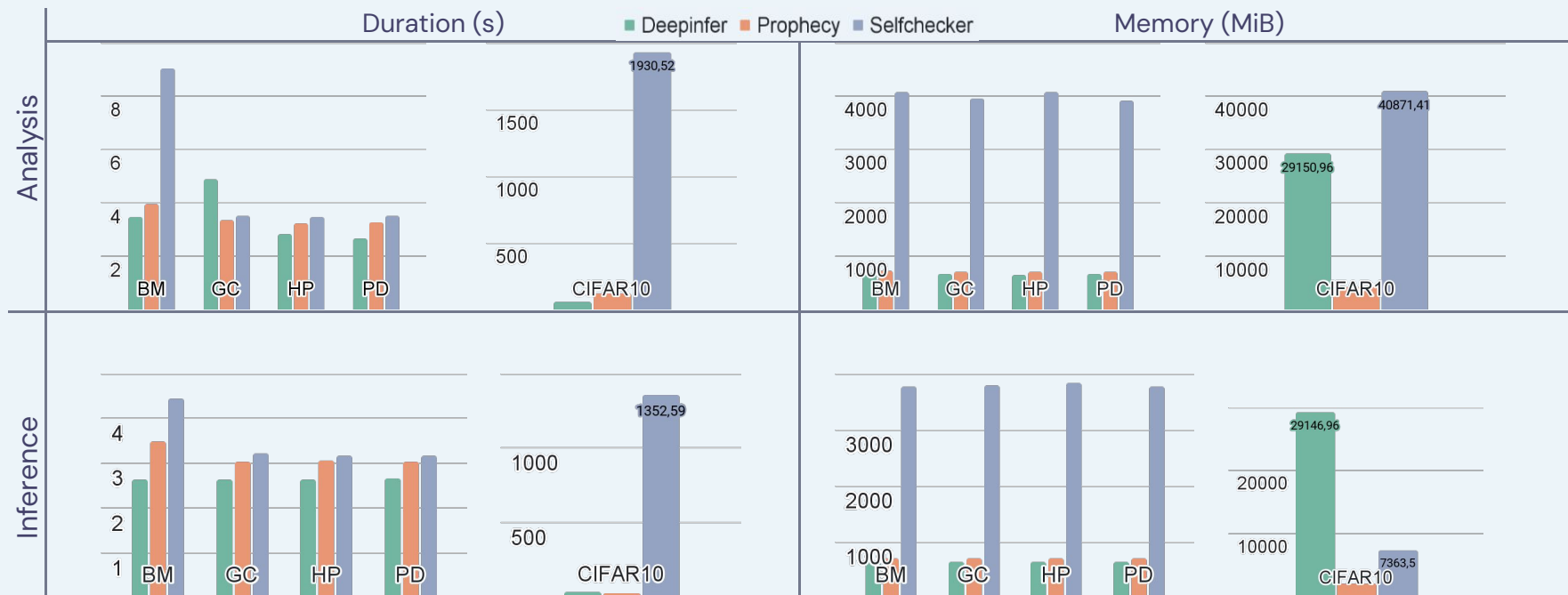
- **DeepInfer** stands out for most of the models;
- **SelfChecker** and **Prophecy** have comparable performance;

Image data:

- **DeepInfer** and **SelfChecker** perform equally well;
- **Prophecy** performs poorly;

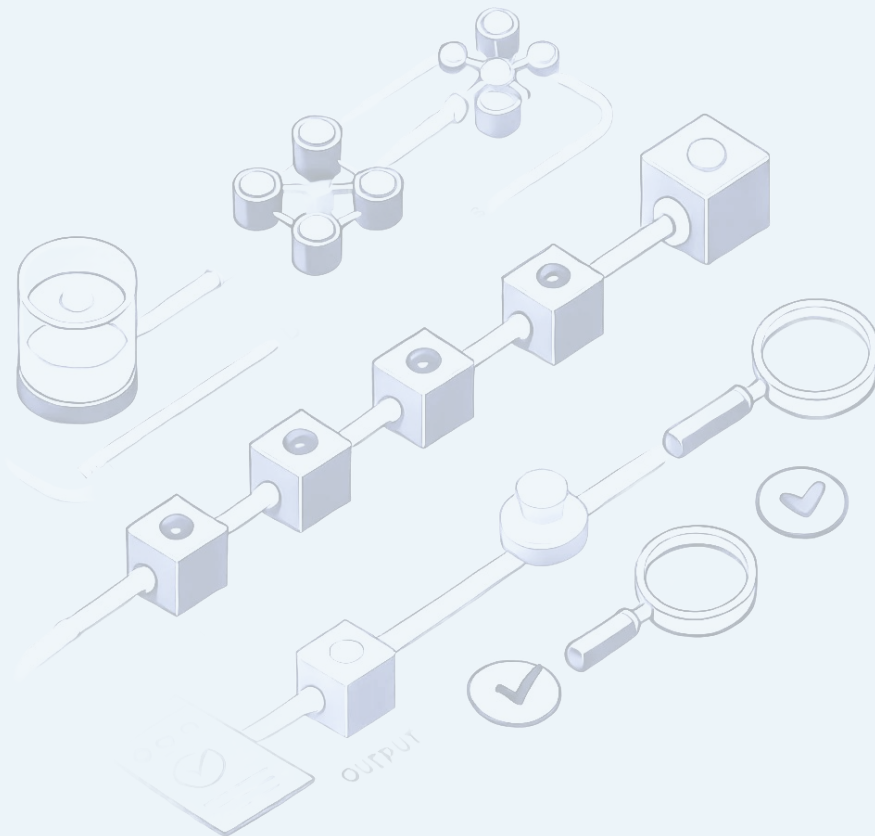
Observation 3: Approaches demonstrate meaningful results with positive MCC values for most models;

Comparison Results – Efficiency



Observation 4: SelfChecker is the most resource intensive while Prophecy is the least across most models;

Conclusion



Lessons Learned



Transparency

- precise inputs;
- proper doc.;
- exact configs.;



Generalizability & Extensibility

- avoid hard-coding;
 - modular code;
- avoid duplicates;



Handle Variability

- multiple trials;
- docker images;

Final Remarks



Evaluated recent DNN
reliability approaches



Hard to reproduce,
replicate, and compare

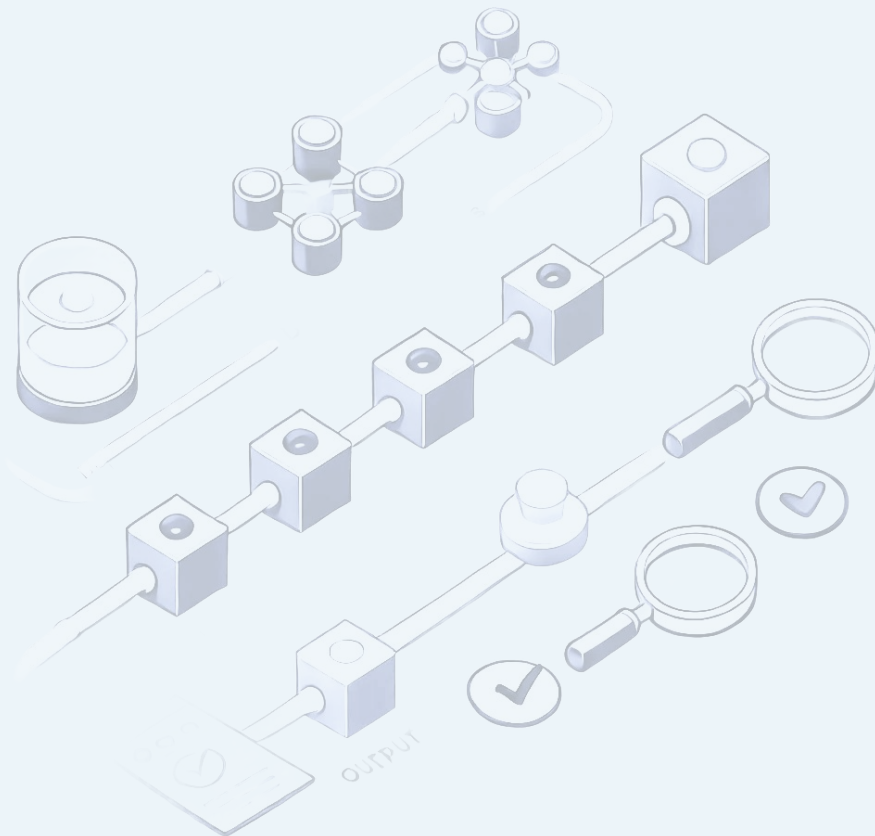


tooling to evaluate
reliability of DNNs

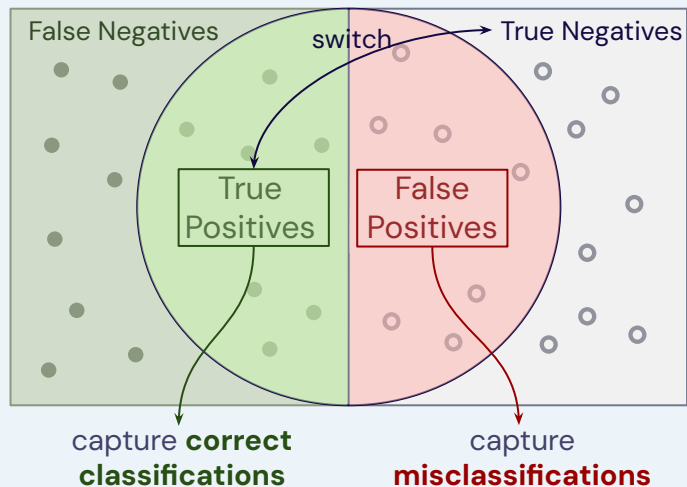
We encourage the community to extend the framework and continue research to obtain sound techniques.

<https://zenodo.org/doi/10.5281/zenodo.12803632>

Questions



Metrics



Conventional Metrics

$$\text{FPR} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$

$$\text{TPR (Recall)} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

used in previous studies

offer valuable insights

sensible to class swapping

considers dist. of elements

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}$$

considers all categories